

Chapter 2 One-Pager — Game Theory and CFR Basics

Research on the possibilities for applying Artificial Intelligence in computer games

Alexander Andreev

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Problem. Chapter 1’s machinery assumes a fully observable state and a stationary environment; poker has neither. Two-player zero-sum imperfect-information games need a different solution concept — Nash equilibrium defined over *information sets* — and a different quality measure: exploitability, not reward. Chapter 2 establishes both, plus the algorithm that computes them, because everything after it is either an approximation of CFR (Chapters 3-5), a deliberate departure from the equilibrium CFR produces (Chapters 7-8), or an attempt to say what equilibrium even means once $N > 2$ (Chapter 9, 11).

Approach. Chance-sampled CFR written from scratch on **Kuhn Poker** — 3 cards, 2 players, 12 information sets: small enough to have a closed-form equilibrium *family* (parameterised by **alpha** in $[0, 1/3]$), rich enough to contain bluffing, mixed strategies and indifference. Hand-coded components: the game engine, recursive counterfactual traversal with regret matching and chance sampling, and an exact exploitability evaluator that enumerates all $2^6 = 64$ pure strategies — because a per-state “oracle” best response is *not* a best response; the best-responder must commit to one action per information set. Trained for 100,000 iterations (seed 0), with an OpenSpiel cross-verification script alongside. **All numbers below are measured** (implementation/step02/models/cfr_results.json).

Key results (measured).

- *CFR lands close to the analytical equilibrium family.* The recovered bluff parameter is **alpha** = 0.1792, and the family’s own internal constraint $P(\text{bet} \mid K) = 3 \cdot \text{alpha}$ holds to within **0.038** (measured **0.5758** against the **0.5377** its own alpha implies); even so, the exploitability of the whole profile is only **0.006**.
- *Pure decisions converge hard; mixed frequencies carry the $O(1/\sqrt{T})$ tail.* King bets and calls at $\geq$ **0.9999**, Jack folds to a bet at **0.99996**, Queen passes at the root at **0.9995** — while the mixing sits **0.005-0.038** off the closed form (Jack bluff after a pass **0.3387** vs $1/3$; Queen call **0.5307** vs $1/3 + \text{alpha} = 0.5126$). The frequencies are the slow part, which is precisely where the residual regret lives.
- *Game value* of the average strategy **-0.0555** against the exact **-1/18 = -0.0556** (the running mean of sampled payoffs over all 100k iterations is **-0.0613**), reproducing Player 0’s structural first-mover disadvantage.
- *The convergence rate is consistent with the theoretical bound.* Log-log exploitability slope **-0.52 $\pm$ 0.02** (5 seeds, 100-100k iterations) against the **-0.5** of the $O(1/\sqrt{T})$ bound.
- *Game value alone is not a validity check.* A strategy that always bluffs the Jack can still average near $-1/18$ while being trivially exploitable; only exploitability catches it. That is why exploitability — not reward — is the metric carried forward into Chapters 7-8 and into the evaluation methodology (Contribution #3).

Thesis connection. Nash is the baseline this thesis exists to leave: Contribution #1 reads a specific opponent in order to justify departing from it, and Contribution #2 bounds how far the departure may go. Concretely, the exact best-response and exploitability code written here becomes the literal oracle reused in Chapters 7-10, and Kuhn remains the smallest testbed where every later claim can be bracketed by an exact answer.

Open questions. Why the *average* strategy converges when the current strategy does not — the intuition (averaging damps the overshoot, as Polyak averaging does) is not a proof. Everything here rests on the two-player zero-sum minimax theorem, and neither CFR’s guarantee nor exploitability’s meaning survives into N-player or general-sum settings (Chapter 9, 11). And even with one sampled deal per iteration, CFR traverses every action sequence of that deal, so this exact algorithm is already out of budget one game up — the motivation for Chapter 3’s Monte Carlo sampling and Chapter 4’s abstraction.